

1.7 Dividing Positive Fractions

Lesson Introduction

 Benchmark	MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions with procedural fluency.		 Learning Target	I can fluently divide positive fractions by positive fractions.
 Benchmark Coherence	Building On MA.5.FR.2.2 Extend previous understanding of multiplication to multiply a fraction by a fraction, including mixed numbers and fractions greater than 1, with procedural reliability.		Working Toward MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.	
 Essential Questions	- How do you divide fractions using the standard algorithm? - How can a tape diagram represent fraction division? - How can you connect models of division and multiplication to the standard algorithm?			
 Lesson Narrative	Tape diagrams are used to model division with fractions before moving to the standard algorithm. Students activate their prior knowledge and use of tape diagrams with division to model the same process for division of fractions. Students begin to formalize their tape model diagrams as standard algorithms.			
 Component Summary	1.7.1 Warm-Up (~5 min.)	Multiplying fractions Error Analysis (<i>SE p. 34</i>)	1.7.3 Guided Practice (10-15 min.)	Practice using any method to divide fractions (<i>SE p. 37</i>)
	1.7.2 Guided Instruction (15-20 min.)	Scaffolded practice dividing fractions with tape diagram and standard algorithm (<i>SE p. 35</i>)	1.7.4 Your Turn! (5-10 min.)	Students apply any method to divide fractions (<i>SE p. 38</i>)
	1.7.5 Wrap-Up (~10 min.)	Practice multiplying and dividing fractions with procedural fluency (<i>SE p. 38</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Reciprocal - Existence of multiplicative inverses <u>Additional Terms:</u> - Mixed number		(Optional) Paper or cardstock (Optional) Markers	If using station rotations or "Scavenger hunt" activity for 1.7.4 Your Turn!, be sure to write each of the problems clearly on a piece of paper or cardstock and place throughout the room before the lesson.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may forget to multiply by the reciprocal or incorrectly find the reciprocal when using the standard algorithm. - Students may confuse or switch the divisor and dividend fractions when computing. - Students may struggle to recognize when they can simplify.	- Consider scaffolding creation of tape diagrams to ensure student work accurately represents division expressions. - During 1.7.2 Guided Instruction and 1.7.3 Guided Practice, make connections between numerators and denominators of factors and how this can be used to recognize when products will be able to be simplified.
	Literacy and Language Routines Think-Pair-Share 1.7.1 Warm-Up positions students to analyze an error made when multiplying mixed numbers. Consider applying Think-Pair-Share routines as students identify an error in the algorithm used, positioning students to recognize and share a common mistake and provide feedback on how to avoid the mistake going forward.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations Students will be working to model fraction division using tape diagrams. This standard emphasizes progressing from modeling problems with diagrams to using algorithms and equations – utilize the progression of the lesson to highlight this progress as students move from the tape diagram to using the standard algorithm.
 Differentiate Instruction	1.7.1 Warm-Up – consider having students write their error analysis notes on a sticky note, including diagrams with labels showing the error and/or correct work, and post these notes in a shared classroom space for students to learn from mistakes and foster a growth mindset in mathematics.	
	1.7.2 Guided Instruction – adjust the tape diagram creation to show fraction division more clearly with color or shading. Consider using colored pencils to circle or shade the dividend into groups that are the size of the divisor.	
	1.7.4 Your Turn! – consider trying ideas like stations, scavenger hunts, problem cards, or relay races to add variety and movement to student practice.	
Lesson Notes:		

1.7.1 Warm-Up: Error Analysis

Component Overview: Students analyze an error made when multiplying fractions and write a brief note to explain the mistake. Consider having students engage in a Think-Pair-Share for this activity.



1.7.1 Warm-Up: Error Analysis

Mateo tried to multiply $3\frac{5}{6} \cdot 4\frac{1}{10}$, but he made an error. His work is shown.

$$3\frac{5}{6} \cdot 4\frac{1}{10} = 12\frac{5}{60} = \left(12\frac{1}{12}\right)$$

1. What error did Mateo make?
He did not convert the mixed fractions to improper fractions prior to multiplying.
2. What is something that Mateo did correctly?
He correctly simplified the fraction.
3. Help Mateo understand and learn from his mistake. Write a brief note to Mateo explaining the mistake he made and how to fix it moving forward.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Answers will vary.



Think-Pair-Share

Consider having students work on the error analysis individually, then share their responses with a partner. Once the partner conversations have taken place, ask a few students to share their thoughts with the class. Also consider having students write on sticky notes to post in a shared classroom space, which could foster a growth mindset and a classroom environment that values learning from mistakes.

4. Find the product of $3\frac{5}{6} \cdot 4\frac{1}{10}$. Show your work.

$$\frac{23}{6} \cdot \frac{41}{10} = \frac{943}{60} = 15\frac{43}{60}$$

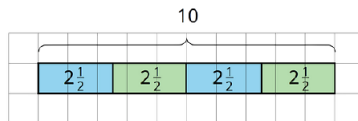
1.7.2 Guided Instruction: Dividing Fractions

Component Overview: Students use a tape diagram to illustrate division and find quotients with fractions. Complete the tape diagrams in the first section as a class to address any misconceptions and build conceptual understanding of dividing fractions. Consider having students work with a partner to apply their understanding to a real-world problem. Finally, students use the standard algorithm to divide fractions. Consider helping students see the reason behind why multiplying by the reciprocal in the standard algorithm is the same as dividing.



1.7.2 Guided Instruction: Dividing Fractions

Recall how tape diagrams can be used to illustrate division.



- Write the division equation represented by the tape diagram.

$$\begin{array}{ccc} \text{dividend} & \div & \text{divisor} & = & \text{quotient} \\ 10 & \div & 2\frac{1}{2} & = & 4 \end{array}$$

- How many groups of $2\frac{1}{2}$ are in 10?
4

- Complete the table by finding each quotient. Use the tape diagram shown.

Tape Diagram	Division Expression	Quotient
	$\frac{1}{2} \div \frac{1}{3}$	$\frac{3}{2}$ or $1\frac{1}{2}$
	$3\frac{1}{4} \div \frac{3}{4}$	$\frac{13}{3}$ or $4\frac{1}{3}$



Throughout this guided instruction, consider using the sentence stem “How many groups of ___ are in ___?” This will reinforce student understanding of the divisor being the group size and the dividend being the total value that is divided into groups.



How many groups of $\frac{1}{3}$ are in $\frac{1}{2}$? How many groups of $\frac{3}{4}$ are in $3\frac{1}{4}$? How can you “see” the quotient to each problem in the tape diagram?



For a visual entry point, consider using colored pencils to show the dividend being divided by the divisor. Use different colors to represent each and circle each group to show how many groups of the divisor are in the dividend.

1.7.2 Guided Instruction: Dividing Fractions (continued)

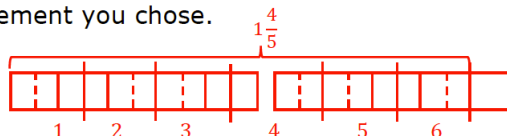
4. Jacquelyn is making ground beef flautas, where each flauta will use $\frac{3}{10}$ pounds of ground beef. She has a total of $1\frac{4}{5}$ pounds of ground beef to use.

- a. Which division statement below represents the number of flautas Jacquelyn can make?

$$\frac{3}{10} \div 1\frac{4}{5}$$

$$1\frac{4}{5} \div \frac{3}{10}$$

- b. Draw a tape diagram to illustrate the division statement you chose.



- c. Solve the division statement.

$$1\frac{4}{5} \div \frac{3}{10} = 6$$

- d. How many $\frac{3}{10}$ pound flautas can she make?

6 flautas

You previously developed an algorithm to use when dividing fractions.



When **dividing fractions**, the division statement can be rewritten as an equivalent multiplication sentence by using the **reciprocal**, or multiplicative inverse, of the second fraction.

$$\frac{3}{4} \div \frac{2}{5} = \frac{3}{4} \times \frac{5}{2} = \frac{15}{8} = 1\frac{7}{8}$$



Existence of multiplicative inverses

For every $a \neq 0$ there exists $\frac{1}{a}$ so that $a \times \frac{1}{a} = 1$ and $\frac{1}{a} \times a = 1$.



Think-Pair-Share

Consider having students complete this real-world fraction division problem using the Think-Pair-Share routine. Be sure to give students individual thinking time before allowing partners to share ideas and answers, then have partners share ideas with the class. Use this opportunity to address any misconceptions when dividing using a tape diagram for the whole class.



For question 4b, how do you use the dividend from your chosen division statement to start creating your tape diagram? How do you know how to divide the tape diagram? For questions 4c and 4d, how can you “see” the quotient in your tape diagram?

1.7.2 Guided Instruction: Dividing Fractions (continued)

With the tape diagrams, you determined $3\frac{1}{4} \div \frac{3}{4} = 4\frac{1}{3}$.

The standard algorithm can also be used to determine the quotient.

Write the given division expression.	$3\frac{1}{4} \div \frac{3}{4}$
Rewrite the expression, rewriting any mixed numbers as fractions greater than 1.	$\frac{13}{4} \div \frac{3}{4}$
Rewrite the division using multiplication by the reciprocal of the second fraction.	$\frac{13}{4} \times \frac{4}{3}$
Multiply the fractions and simplify.	$\frac{52}{12} = \frac{13}{3} = 4\frac{1}{3}$

5. Find the quotient $5\frac{2}{3} \div 2\frac{4}{9}$ using the algorithm.

$$\frac{17}{3} \div \frac{22}{9} = \frac{17}{3} \cdot \frac{9}{22} = \frac{153}{66} = \frac{51}{22} \text{ or } 2\frac{7}{22}$$



Encourage students to formulate their own steps, then discuss the steps in the table. Relate the steps to the tape diagram.



Consider using the algorithm as an opportunity to dive deeper into why multiplying by the reciprocal is the same as dividing. For example, provide fraction bar or tape diagram evidence that shows $10 \div 2 = 5$, and $10 \times \frac{1}{2} = 5$.

1.7.3 Guided Practice: Dividing Fractions

Component Overview: Students apply any method to divide fractions. The table in this section summarizes the important skills of rewriting mixed numbers as fractions before computing, writing the equivalent multiplication sentence with the reciprocal, and simplifying the quotient if possible. Consider giving students time to work individually, then checking responses as a class to confirm understanding and correct usage of the algorithm.



1.7.3 Guided Practice: Dividing Fractions

1. Complete the table to find each quotient.

Division Sentence	Rewritten as Fractions (If needed)	Equivalent Multiplication Sentence	Quotient
$\frac{7}{8} \div 2\frac{3}{10}$	$\frac{7}{8} \div \frac{23}{10}$	$\frac{7}{8} \cdot \frac{10}{23}$	$\frac{70}{184} = \frac{35}{92}$
$\frac{2}{7} \div \frac{3}{4}$		$\frac{2}{7} \cdot \frac{4}{3}$	$\frac{8}{21}$
$8\frac{2}{3} \div \frac{5}{6}$	$\frac{26}{3} \div \frac{5}{6}$	$\frac{26}{3} \cdot \frac{6}{5}$	$\frac{156}{15} = \frac{52}{5}$ or $10\frac{2}{5}$
$4\frac{4}{5} \div 1\frac{1}{2}$	$\frac{24}{5} \div \frac{3}{2}$	$\frac{24}{5} \cdot \frac{2}{3}$	$\frac{48}{15} = \frac{16}{5}$ or $3\frac{1}{5}$

2. Find each quotient using your preferred method. Simplify when possible.

a. $\frac{3}{4} \div \frac{9}{14}$

$$\begin{aligned} \frac{3}{4} \cdot \frac{14}{9} \\ = \frac{42}{36} \\ = \frac{7}{6} \text{ or } 1\frac{1}{6} \end{aligned}$$

b. $6\frac{7}{8} \div \frac{5}{7}$

$$\begin{aligned} \frac{55}{8} \div \frac{5}{7} \\ = \frac{55}{8} \cdot \frac{7}{5} \\ = \frac{385}{40} \\ = \frac{77}{8} \text{ or } 9\frac{5}{8} \end{aligned}$$



Consider highlighting products in this practice that can be simplified, like the bottom two rows of the table in question 1 and both parts of question 2. Direct student attention to fractions whose numerator and denominator share a common factor to help them recognize when products can be simplified.



Consider asking students:

- What do you notice about the numerator and denominator in the product?
- How do you know a fraction can be simplified?
- Why do we simplify fractions and mixed numbers?

1.7.4 Your Turn!

Component Overview: Students practice finding quotients and simplifying when possible. Consider having students work in partners and using station rotations or having a “scavenger hunt” to search for and solve each problem to switch up procedural practice. If using station rotations or the “scavenger hunt,” be sure to write each problem below on a piece of paper and place in various locations around the classroom before beginning.



1.7.4 Your Turn!

1. Find each quotient. Simplify when possible.

a. $\frac{1}{18} \div \frac{1}{9}$

$$\begin{aligned} & \frac{1}{18} \cdot \frac{9}{1} \\ &= \frac{9}{18} \\ &= \frac{1}{2} \end{aligned}$$

b. $2\frac{2}{5} \div \frac{1}{10}$

$$\begin{aligned} & \frac{12}{5} \div \frac{1}{10} \\ &= \frac{12}{5} \cdot \frac{10}{1} \\ &= \frac{120}{5} \\ &= 24 \end{aligned}$$

c. $\frac{3}{15} \div 6$

$$\begin{aligned} & \frac{3}{15} \div \frac{6}{1} \\ &= \frac{3}{15} \cdot \frac{1}{6} \\ &= \frac{3}{90} \\ &= \frac{1}{30} \end{aligned}$$

d. $9 \div \frac{3}{2}$

$$\begin{aligned} & \frac{9}{1} \div \frac{3}{2} \\ &= \frac{9}{1} \cdot \frac{2}{3} \\ &= \frac{18}{3} \\ &= 6 \end{aligned}$$

e. $1\frac{5}{9} \div 5\frac{5}{8}$

$$\begin{aligned} & \frac{14}{9} \div \frac{45}{8} \\ &= \frac{14}{9} \cdot \frac{8}{45} \\ &= \frac{112}{405} \end{aligned}$$

f. $\frac{1}{20} \div \frac{1}{4}$

$$\begin{aligned} & \frac{1}{20} \div \frac{1}{4} \\ &= \frac{1}{20} \cdot \frac{4}{1} \\ &= \frac{4}{20} \\ &= \frac{1}{5} \end{aligned}$$

g. $\frac{4}{12} \div \frac{1}{7}$

$$\begin{aligned} & \frac{4}{12} \div \frac{1}{7} \\ &= \frac{28}{12} \\ &= \frac{14}{6} \text{ or } 2\frac{1}{3} \end{aligned}$$

h. $2 \div 3\frac{4}{15}$

$$\begin{aligned} & \frac{2}{1} \div \frac{49}{15} \\ &= \frac{2}{1} \cdot \frac{15}{49} \\ &= \frac{30}{49} \end{aligned}$$

i. $1\frac{6}{10} \div 5\frac{3}{4}$

$$\begin{aligned} & \frac{16}{10} \div \frac{23}{4} \\ &= \frac{16}{10} \cdot \frac{4}{23} \\ &= \frac{64}{230} \\ &= \frac{32}{115} \end{aligned}$$



Consider a variety of methods for implementation.

- **Scavenger or Stations:** Station rotation or “scavenger hunt” activity with a card for each problem hidden around the room for students to find and solve.
- **Student Choice:** Students opting in to self-select quotients to find, increasing student choice and providing important observational feedback for which problems students choose and why.
- **Task Cards:** Print and sort problems into Task Cards using the categories of improper, mixed, or other types of fractions and have students practice by category.
- **Trashketball:** Student favorite – each correct response earns a chance to shoot a ball of paper into the can. Most made baskets means a homework pass!

1.7.5 Wrap-Up: Ticket Out the Door

Component Overview: Consider using all or parts of this exercise as a formative assessment to gather information on student learning from this lesson and the last lesson. This Wrap-Up assesses students' ability to both multiply and divide fractions using any method and simplifying when possible. Students should not use a calculator to complete the Wrap-Up.



1.7.5 Wrap-Up: Ticket Out the Door

1. Find each product or quotient. Simplify when possible.

a. $\frac{2}{11} \div \frac{1}{3}$

$$\frac{2}{11} \cdot \frac{3}{1} = \frac{6}{11}$$

b. $\left(\frac{2}{23}\right)\left(3\frac{3}{23}\right)$

$$\frac{2}{23} \cdot \frac{72}{23} = \frac{144}{529}$$



Consider adding complexity by having students solve one or more of the problems in two ways: with the standard algorithm and with a diagram or model. For dividing fractions, students can use a tape diagram as modeled in this lesson. For multiplying fractions, students can use an area model.

c. $2\frac{27}{32} \times \frac{3}{182}$

$$\frac{91}{32} \cdot \frac{3}{182} = \frac{3}{64}$$

d. $3\frac{3}{4} \div 4\frac{1}{6}$

$$\begin{aligned} \frac{15}{4} \div \frac{25}{6} &= \frac{15}{4} \cdot \frac{6}{25} \\ &= \frac{90}{100} \\ &= \frac{9}{10} \end{aligned}$$